

Public Economics (ECON 131)

Section #8: Externalities

GSI: Steve Ramos

Contents

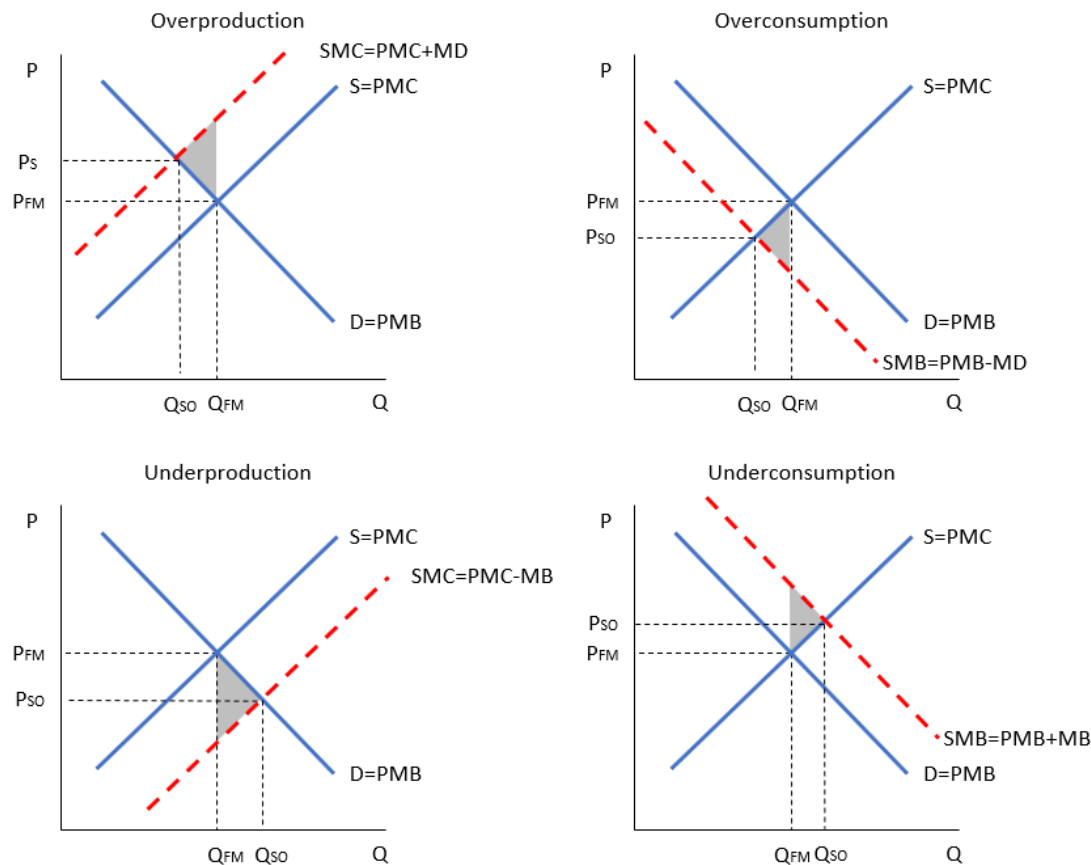
	Page
1 Externalities	1
1.1 Key concepts	1
1.2 Practice Problems	3
1.2.1 Based on Gruber Ch.05, Q.13	3
1.3 Additional Practice Problems	5
1.3.1 Based on Gruber Ch.05, Q.14	5
1.3.2 Gruber Ch.05, Q.4	6

1 Externalities

1.1 Key concepts

- **Market failure:** A problem that causes the market economy to deliver an outcome that does not maximize efficiency.
 - Externalities, asymmetric information, individual failures
- **Externality:** An externality arises when the action of an agent directly makes other agents better off (positive externality) or worse off (negative externality), with no costs or benefits to the acting agent.
- Externalities can occur because of production or consumption
- Private vs. social marginal cost:
 - Private marginal cost (PMC): is the direct marginal cost of producing/consuming an additional unit of a good
 - Social marginal cost (SMC): is the PMC + marginal damages (MD), ie the PMC + any additional costs associated with production/consumption that are imposed on others in the economy

- In a free market, equilibrium where $PMB = PMC$
- Social optimum at $SMB = SMC$
- If $PMB \neq SMB$ and/or $PMC \neq SMC \Rightarrow$ private market leads to inefficient outcome



- We say an agent **internalizes the externality** when, due either to private negotiations or government actions, the price of an action fully reflects the external costs or benefits of that action.
 - Private negotiations can bring result in socially optimal quantity as long as (a) there are well-defined property rights (b) costless bargaining, and (c) competitive markets (**Coase theorem**—but see lecture slides for some limitations)
 - Public policy can enter when Coase conditions are not met:
 - * price policy (tax or subsidy)
 - * quantity regulation (command and control or tradable permits)

1.2 Practice Problems

1.2.1 Based on Gruber Ch.05, Q.13

Suppose that demand for a product is $Q = 1,000 - 4P$ and supply is $Q = -200 + 2P$. Furthermore, suppose that the marginal external damage of this product is \$6 per unit.

- (a) How many more units of this product will the free market produce than is socially optimal?
- (b) Calculate the deadweight loss associated with the externality.

Solution:

- (a) To answer this question, first calculate what the free market would do by setting demand equal to supply:

$$1,000 - 4P = -200 + 2P \quad \text{or} \quad 1,200 = 6P.$$

Thus, $P_{FM} = 200$ and $Q_{FM} = 1,200 - 4(200) = 200$.

The socially optimal level occurs when the marginal external cost is included in the calculation. I will list a two different (but equivalent) ways to solve this problem:

- i) Externality price added to consumers (i.e. *overconsumption*):

Suppose the \$6 externality were added to the price each consumer had to pay (as if there were a tax on consumers). Then, the consumer pays \$6 more than the producer receives, so $P^D = P^S + 6$, where P^D is the price that consumers pay, and P^S is the price that producers receive.

Then demand would be $Q^D = 1,000 - 4P^D = 1,000 - 4(P^S + 6)$. Supply is still $Q^S = -200 + 2P^S$.

Solving for P^S and Q by setting demand equal to supply:

$$1,000 - 4(P^S + 6) = -200 + 2P^S$$

Thus, $P_{SO}^S = 196$. To find Q_{SO} , plug this price back into the supply curve (since we are plugging in the “supply” price) to get $Q_{SO} = 192$.

- ii) Externality price added to producers (i.e. *overproduction*):

Suppose the \$6 externality were subtracted from the price each producer received (as if there were a tax on producers). Then, the producer receives \$6 less than the consumer pays, so $P^S = P^D - 6$.

Then, supply would be $Q^S = -200 + 2P^S = -200 + 2(P^D - 6)$ and demand would still be $Q^D = 1,000 - 4P^D$.

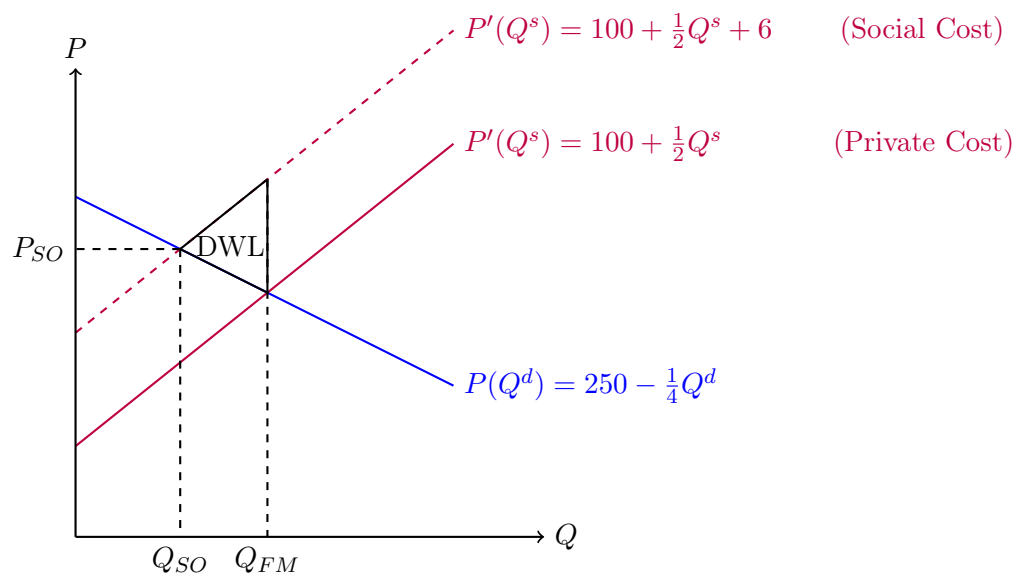
Solving for P^D and Q by setting demand equal to supply:

$$-200 + 2(P^D - 6) = 1,000 - 4P^D$$

Thus, $P_{SO}^D = 202$. Plugging the price back into the demand curve (since we are plugging in the “demand” price), we get $Q_{SO} = 192$.

Note that either solution gives $Q_{SO} = 192$. This means that the free market will produce 8 units more than is socially optimal.

The results for the case of overproduction are also listed in the graph below.



- (b) Deadweight loss is the area of the triangle labelled on the graph. The height of the triangle is 6 (the amount of the marginal damage) and width is 8 (i.e. $Q_{FM} - Q_{SO}$) : $\frac{1}{2}(6 \times 8) = 24$.

1.3 Additional Practice Problems

1.3.1 Based on Gruber Ch.05, Q.14

The marginal damage averted from pollution cleanup is $MD = 200 - 5Q$. The marginal cost associated with pollution cleanup is $MC = 20 + Q$.

- (a) What is the optimal level of pollution reduction?
- (b) Show that this level of pollution reduction could be accomplished through taxation. What tax per unit would generate the optimal amount of pollution reduction?

Solution:

- (a) Here we are asked to find the quantity where $SMB = SMC$. The damage averted is the SMB . The cost of cleanup faced by the private polluter is the PMC and the SMC , so:

$$200 - 5Q = 20 + Q \quad \text{or} \quad Q = 30$$

- (b) We want to use a tax to get the private polluter to clean up the optimal amount of pollution, 30. By setting a tax T on each unit of pollution, the government will induce the polluters to clean it up as long as the marginal cost of cleanup (PMC) is less than or equal to the tax.

So the total amount of pollution cleanup for a given tax will solve:

$$20 + Q^* = T$$

To implement the social optimum of 30 units of pollution thus requires a tax of $20 + 30 = 50$.

1.3.2 Gruber Ch.05, Q.4

In the midwestern United States, where winds tend to blow from west to east, states tend to more easily approve new polluting industries near their eastern borders than in other parts of the state. Why do you think this is true?

Solution:

- When a state approves new polluting industries, it imposes an externality on neighboring "downwind" states.
- The polluting state does not face the marginal external damage caused to citizens of the "downwind" states.
- This is an example of the case of overproduction – the state (and polluters) don't internalize the cost of the pollution, and so they produce more than socially optimal.
- As a counterexample, there may be other reasons that these industries are located on the eastern borders. Lots of cities are on the east borders of midwestern states, likely because there was a gradual process of westward expansion of settlers during the 19th century. Examples include Detroit, Chicago, Milwaukee, and the Twin Cities. It may be that industrial plants tend to be located near such population (employment) centers.